ALGEBRAIC STACKS WITHOUT AUTOMORPHISMS ARE EXACTLY THOSE REPRESENTABLE BY AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span. Same-chapter inputs not carded are declared external.

Allowed standard prerequisites (editorial).

- Setoid characterizations. Stacks, lemmas characterize-stack-in-setoids and stack-in-setoids-characterize (cross-chapter, named). Give (1) iff (2) and the equivalence j: X -> S_F.
- DM no-automorphisms lemma. lemma-algebraic-stack-no-automorphisms (carded L26683). Gives (3) implies (1), as cited in the source proof.
- Diagonal transfer to presheaves. lemma-map-presheaves-representable-by-algebraic-spaces (same chapter, NOT carded, declared external). Transfers representability of Delta_X to Delta_F.
- Setoid property transfer. lemma-map-fibred-setoids-property (same chapter, NOT carded, declared external). Concludes h_U -> F is surjective smooth.
- Smooth implies flat and lfp. Morphisms of Spaces, lemmas smooth-locally-finite-presentation and smooth-flat (cross-chapter, named). Feeds the property-implication lemma.
- Property implication. lemma-representable-transformations-property-implication (same chapter, NOT carded, declared external). Deduces flatness and lfp of h_U -> F.
- Final bootstrap theorem. Bootstrap, theorem final-bootstrap (cross-chapter, named). Final step: F is an algebraic space.

Source reference. Original reference 03Y2: focused-seeds/geometry/S9-alg-alg-stack-no-automorphisms/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04S6: focused-seeds/geometry/S9-alg-alg-stack-no-automorphisms/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045H: sources/stacks-algebraic/pinned-original-source.tex, source lines 1730–1730. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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Source reference. Original lemma 04TC: sources/stacks-algebraic/pinned-original-source.tex, source lines 1138–1138. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0458: sources/stacks-algebraic/pinned-original-source.tex, source lines 839–839. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045F: sources/stacks-algebraic/pinned-original-source.tex, source lines 1264–1264. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04TA: focused-seeds/geometry/S9-alg-alg-stack-no-automorphisms/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04AJ: focused-seeds/geometry/S9-alg-alg-stack-no-automorphisms/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04ZM: focused-seeds/geometry/S9-alg-alg-stack-no-automorphisms/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0432: focused-seeds/geometry/S9-alg-alg-stack-no-automorphisms/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{X}$ be an algebraic stack over S. The following are equivalent*

(1) *$\mathcal{X}$ is a stack in setoids,*
(2) *the canonical 1-morphism $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is an equivalence, and*
(3) *$\mathcal{X}$ is representable by an algebraic space.*

Proof. The equivalence of (1) and (2) follows from Stacks, Lemma 04ZM. The implication (3) $\Rightarrow$ (1) follows from Lemma 045H. Finally, assume (1). By Stacks, Lemma

0432 there exists an equivalence $j : \mathcal{X} \to \mathcal{S}_F$ where F is a sheaf on $(Sch/S)_{fppf}$. By Lemma 0458 the fact that $\Delta_{\mathcal{X}}$ is representable by algebraic spaces, means that $\Delta_F : F \to F \times F$ is representable by algebraic spaces. Let U be a scheme and let $x : (Sch/U)_{fppf} \to \mathcal{X}$ be a surjective smooth morphism. The composition $j \circ x : (Sch/U)_{fppf} \to \mathcal{S}_F$ corresponds to a morphism $h_U \to F$ of sheaves. By Bootstrap, Lemma 03Y2 this morphism is representable by algebraic spaces. Hence by Lemma 04TC we conclude that $h_U \to F$ is surjective and smooth. In particular it is surjective, flat and locally of finite presentation (by Lemma 045F and the fact that a smooth morphism of algebraic spaces is flat and locally of finite presentation, see Morphisms of Spaces, Lemmas 04AJ and 04TA). Finally, we apply Bootstrap, Theorem 04S6 to see that F is an algebraic space. □